COMPSCI5079(M)
Cryptography & Secure Development

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Lecture 2:

Traditional Encryption and Decryption schemes

Lecture Outline

1. Types of Attack
   - Threats that need to be protected against.
2. Transposition and substitution ciphers.
3. Homophonic Ciphers
   - Plaintext letters can have several ciphertext letters.
4. Polyalphabet Ciphers and Rotor Machines.
   - Mechanical encryption devices.
5. Running Key Ciphers
   - The key is the same length as the plaintext

Types of Attack

Traditional Encryptions

- Covers the time period up to the invention of the electronic computer.

- All traditional encryption schemes are single-key systems, with two main variations.
  - *Transposition ciphers*, where the characters are rearranged.
  - *Substitution ciphers*, where the characters are substituted.

- Transposition and substitution can be combined.

Types of Attack

- Attempts to break the code normally depend on the information available.

1. Ciphertext Only

2. Known plaintext
   - Know part of the message (i.e., plaintext) as well as the ciphertext and use this information to find the key.

3. Chosen plaintext
   - Plant some plaintext and examine the ciphertext to find the key.

Types of Attack: Methods

- Brute force (exhaustive search)
  - Try all possible keys.
- Letter frequency.
  - Use the letter frequencies in the ciphertext.
- Di-gram and tri-gram frequencies
  - Frequencies of pairs and triples of characters in the ciphertext.

- The following slide shows the letter frequencies: totals and occurrences per 1000 letters from the file `dracula.txt.`
- Some letters are much more frequent than others.

Letters Statistic in dracular.txt

a ---	52337	82	n ---	43597	68
b ---	8987	14	o ---	50331	79
c ---	13516	21	p ---	9158	14
d ---	28539	45	q ---	625	1
e ---	79302	124	r ---	34951	55
f ---	13991	22	s ---	39484	62
g ---	12670	20	t ---	58123	91
h ---	43201	68	u ---	17923	28
i ---	42602	67	v ---	5871	9
j ---	813	1	w ---	18057	28
k ---	6201	10	x ---	781	1
l ---	26115	41	y ---	12671	20
m ---	17758	28	z ---	351	1

Confusion and Diffusion

- A good encryption system will try to make sure that there is little relationship between plaintext letters, the key, and ciphertext letters.
- In particular, **small changes** to the plaintext or the key should produce **large changes** to the ciphertext.
- An encryption system has *good confusion* if changing *one bit* in the key changes roughly *half the bits* of the ciphertext.
- An encryption system has *good diffusion* if changing *one bit* in the plaintext changes roughly *half the bits* of the ciphertext.